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MATHEMATICS EXTENSION 2

ASSESSMENT TASK NO 2

Date 20 March 2014

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen Black pen is preferred.
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/ or calculations

Total marks – 100

Section 1 Pages 2 - 5

10 marks

- Attempt Questions 1 – 10
- Allowed about 15 minutes for this section

Section 2 Pages 6 - 13

90 marks

- Attempt Questions 11- 16
- Allow about 2 hours and 45 minutes for this section

M.C.	/10
Q11	/15
Q12	/15
Q13	/15
Q14	/16
Q15	/14
Q16	/15

This assessment task constitutes 30% of the Higher School Certificate Course Assessment.

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

Multiple choice questions (10 marks).**Marks**

Allow 15 minutes for this section

Use the multiple-choice answer sheet for Q1 to 10

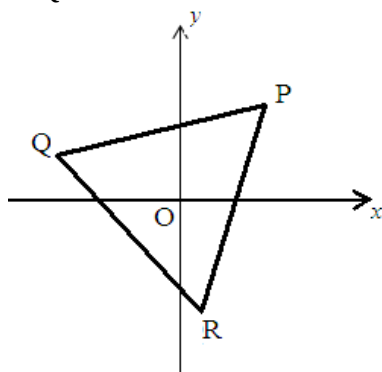
1. $\sqrt{3} + i$ expressed in modulus argument form is:

- (A) $2 \operatorname{cis} \frac{\pi}{4}$ (B) $2 \operatorname{cis} \frac{\pi}{6}$
(C) $\sqrt{2} \operatorname{cis} \frac{\pi}{3}$ (D) $2 \operatorname{cis} \pi$

2. The parametric equations of the hyperbola $\frac{(x+1)^2}{4} - \frac{(y-3)^2}{2} = 1$ are:

- (A) $x = -1 + 2 \sec \theta, y = 3 + \sqrt{2} \tan \theta$
(B) $x = 1 + 2 \sec \theta, y = 3 - \sqrt{2} \tan \theta$
(C) $x = 2 \sec(\theta - 1), y = \sqrt{2} \tan(\theta + 3)$
(D) $x = 3 + 2 \sec \theta, y = -1 + \sqrt{2} \tan \theta$

3. P represent the complex number $p = 1 + 2i$ on an Argand diagram. Given that PQR is an equilateral triangle whose centre is at the origin O. Find the complex number q , represented by the point Q.



- (A) $\frac{-(1+2\sqrt{3})+i(\sqrt{3}-2)}{2}$ (B) $\frac{(1+2\sqrt{3})-i(\sqrt{3}-2)}{2}$
(C) $\frac{-(1+3\sqrt{2})+i(\sqrt{3}-3i)}{2}$ (D) $\frac{-(1+2\sqrt{3})+i(\sqrt{3}-2i)}{4}$

4. Which pair of equations gives the asymptotes of $9x^2 - 16y^2 = 144$

(A) $y = \pm \frac{4x}{3}$

(B) $y = \pm \frac{4}{3}$

(C) $y = \pm \frac{9x}{16}$

(D) $y = \pm \frac{3x}{4}$

5. Factorize $x^4 + x^2 - 12$ over the complex field.

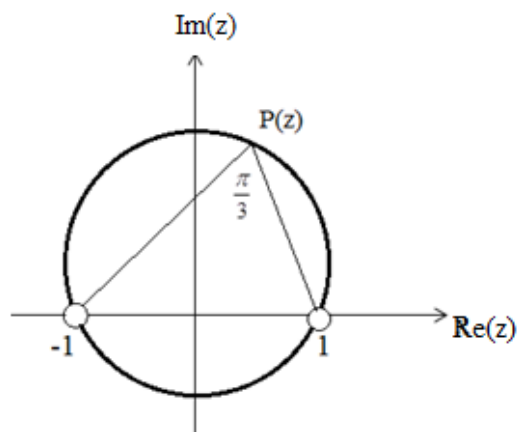
(A) $(x^2 + 4)(x - i\sqrt{3})(x + i\sqrt{3})$

(B) $(x - 2i)(x - \sqrt{3})(x + \sqrt{3})$

(C) $(x + 2i)(x - 2i)(x + \sqrt{3})(x - \sqrt{3})$

(D) $(x + 2)(x - 2)(x + i\sqrt{3})(x - i\sqrt{3})$

6. The locus of z is represented by the circle in the Argand diagram below. The centre and radius of the circle are:



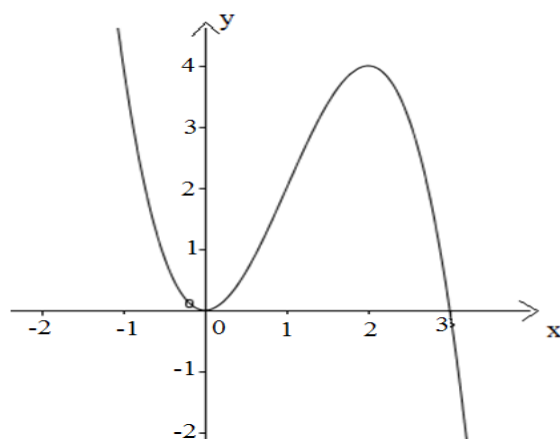
(A) $C\left(0, \frac{\sqrt{3}}{2}\right)$ and $r = \frac{\sqrt{3}}{2}$

(B) $C\left(0, \frac{\sqrt{3}}{3}\right)$ and $r = \frac{\sqrt{3}}{2}$

(C) $C\left(0, \frac{\sqrt{3}}{3}\right)$ and $r = \frac{2\sqrt{3}}{3}$

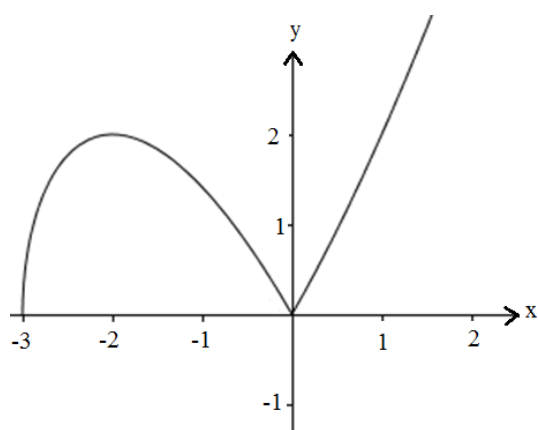
(D) $C(0, \sqrt{3})$ and $r = \frac{2\sqrt{3}}{3}$

7. The sketch of $y = f(x)$ is shown.

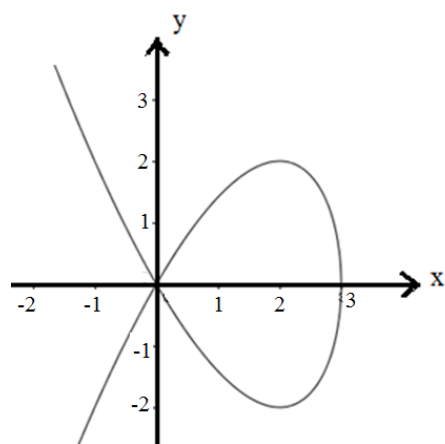


The graph which represents $y^2 = f(x)$ is:

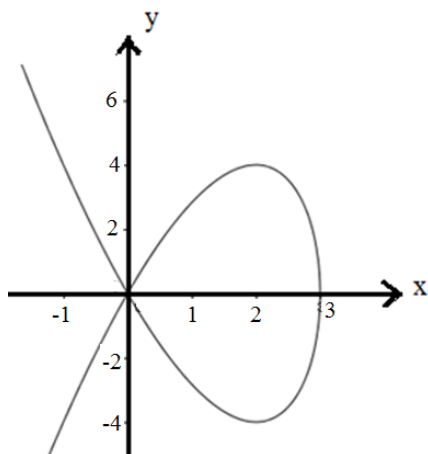
(A)



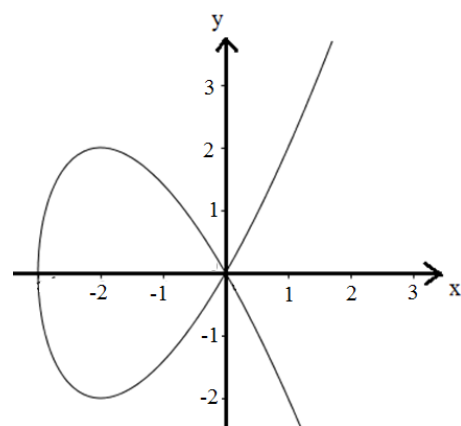
(B)



(C)



(D)



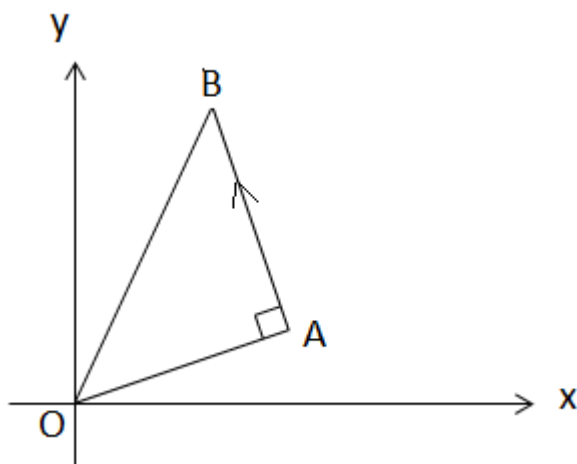
8. The length of the latus rectum of the hyperbola $\frac{x^2}{25} - \frac{y^2}{4} = 1$ is:
- (A) $\frac{8}{25}$
- (B) $-\frac{8}{25}$
- (C) $\frac{4}{25}$
- (D) $\frac{8}{5}$
9. Given $4x^3 + 3x - 2$ has a rational zero, find all other zeros.
- (A) $\frac{1}{2}$ and $\pm \frac{1 + \sqrt{15}i}{4}$
- (B) $\frac{1}{2}$ and $\frac{-1 \pm \sqrt{15}i}{2}$
- (C) $\frac{1}{2}$ and $\frac{-1 \pm i\sqrt{15}}{4}$
- (D) $\frac{1}{2}$ and $\frac{1 \pm i\sqrt{15}}{4}$
10. If α, β and λ are the roots of the equation $x^3 + 2x^2 - 5x - 6 = 0$, the value of $(\alpha - 1)(\beta - 1)(\lambda - 1)$ is:
- (A) 12
- (B) - 6
- (C) 15
- (D) 8

(a) Evaluate $|3 - 2i|$ 1

(b) Express in the form $x + iy$, where x and y are real

$$(3 + 4i)(4 - i) \quad \text{2}$$

(c)



In the Argand diagram $\triangle OAB$ is isosceles and right angled at A. $\vec{AB}$ represents the complex number α

(i) What complex number corresponds to the vertex A? 1

(ii) What complex number correspond to the vertex B? 1

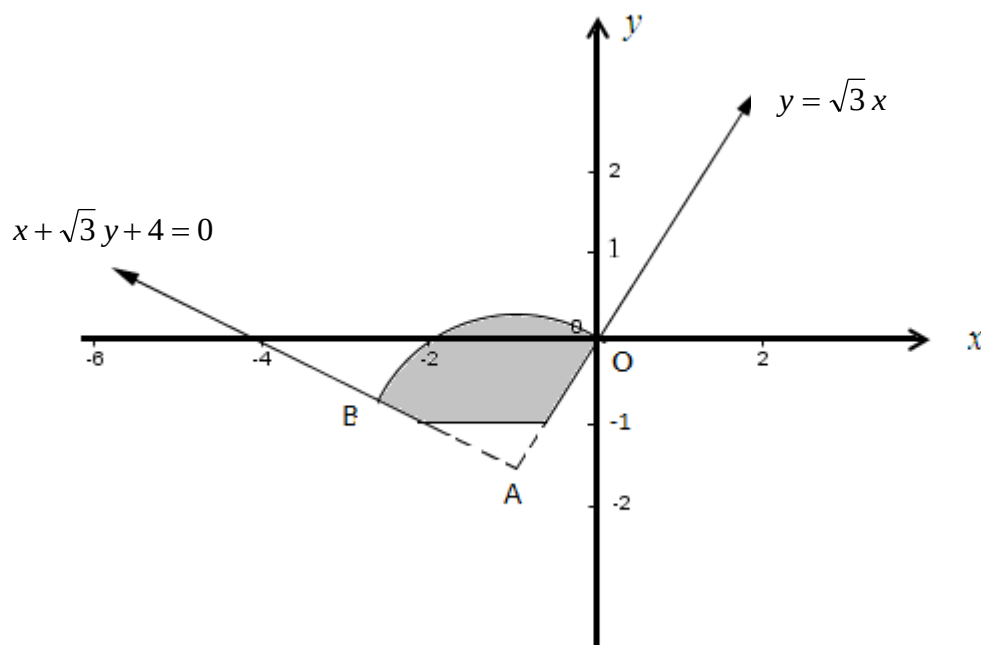
(iii) Show that the area of $\triangle OAB$ is $\frac{1}{2} \alpha \bar{\alpha}$. 1

Question 11 continues on page 7

- (d) The complex number z is given by:

$$z = 1 + \frac{1+i}{1-i}$$

- (i) Express z in modulus-argument form. 2
- (ii) Hence find z^5 in the form $x + iy$, where x and y are real. 2
- (e) In the Argand diagram below the shaded region is a part of a quarter of a circle whose centre is A and its radius is AO . Find the conditions that should be satisfied by the complex numbers Z which represents the shaded region. 5



End of Question 11

- (a) The equation $2x^3 - 2x^2 + 1 = 0$ has roots α , β and γ . **3**

Find the equation with roots α^3 , β^3 and γ^3 .

- (b) The graph of function $f(x) = x\sqrt{9 - x^2}$ is given below (on page 9).

- α) Find the co-ordinates of the stationary points and x-intercepts. **3**

Then copy or trace the graph into your writing booklet and mark all the important features.

- β) Use the graph of $f(x)$ to sketch each of the following on a separate number plane

- (i) $y = f(-x)$ **1**

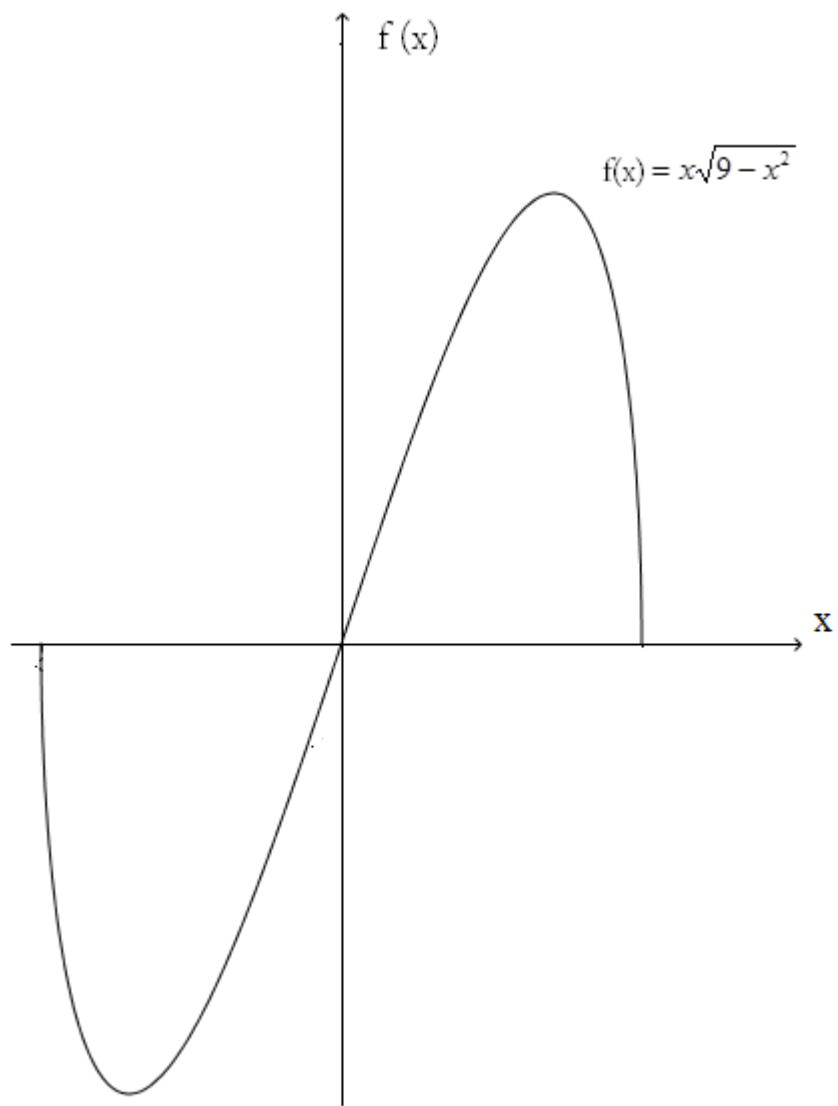
- (ii) $y = \frac{1}{f(x)}$ **2**

- (iii) $y = |f(x)|$ **1**

- (iv) $y = (f(x))^2$ **2**

- (v) $y = e^{f(x)}$ **3**

Question 12 continues on page 9



End of Question 12

-
- (a) Find the equation of the locus of a point $P(x, y)$ such that its distance from the point $S(ae, 0)$ to the line $x = \frac{a}{e}$ is in the ratio $e : 1$ where $e < 1$. **3**
- (b) For the hyperbola $5x^2 - 4y^2 = 20$:
- (i) Find the equations of the asymptotes **1**
- (ii) If the point $P(2 \sec \theta, \sqrt{5} \tan \theta)$ lies on this hyperbola prove that the equation of the tangent to the hyperbola at P is $\frac{x \sec \theta}{2} - \frac{y \tan \theta}{\sqrt{5}} = 1$. **3**
- (iii) If the tangent cuts the asymptotes at L and M, prove that $LP = PM$. **4**
- (iv) Prove that the area of triangle OLM is independent of the position of P, **4**
where O is the origin.

End of Question 13.

-
- (a) The polynomial $P(z)$ has the equation $P(z) = z^4 - 5z^3 + 7z^2 + 3z - 10$ **2**
- Given that $z - 2 + i$ is a factor of $P(z)$, express $P(z)$ as a product of factors with real coefficients.
- (b) Let $f(t) = t^3 + ct + d$, where c and d are constants.
- Suppose that the equation $f(t) = 0$ has three distinct real roots, t_1 , t_2 and t_3 .
- (i) Find $t_1 + t_2 + t_3$. **1**
- (ii) Show that $t_1^2 + t_2^2 + t_3^2 = -2c$. **2**
- (iii) Since the roots are real and distinct, the graph of $y = f(t)$ has two turning points, at $t = u$ and $t = v$, and $f(u) \cdot f(v) < 0$. **4**
- Show that $27d^2 + 4c^3 < 0$.
- (c)
- (i) Suppose that k is a double root of the polynomial equation $f(x) = 0$. Show that $f'(k) = 0$. **2**
- (ii) The polynomial $P(x) = ax^7 + bx^6 + 1$ is divisible by $(x-1)^2$. Find the coefficient a and b . **2**
- (iii) Let $E(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$. Prove $E(x) = 0$ has no double root **2**

End of Question 14

- (a) The ellipse E, is given in terms of the complex number z by: 2

$$|z + 3| + |z - 3| = 10. \text{ Find the Cartesian equation of E.}$$

- (b) Find the integral $\int \frac{x-1}{\sqrt{2x+3}} dx$ 2

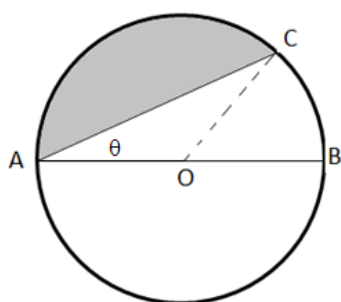
- (c) (i) Show that the tangent to the rectangular hyperbola $xy = c^2$ at the point $T\left(ct, \frac{c}{t}\right)$ has equation $x + t^2y = 2ct$. 2

- (ii) The tangents to the rectangular hyperbola $xy = c^2$ at the points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, where $pq = 1$, intersect at R.

- α) Find the equation of the locus of R. 3

- β) State any restriction on the values of x for this locus. 1

- (d)



In the diagram, AB is the diameter of the circle, and $\angle BAC = \theta$.

- (i) If the area of the shaded region is one third of the area of the circle, show that $\sin 2\theta = \frac{\pi}{3} - 2\theta$ 3

- (ii) Show graphically that this equation has only one solution. 2

End of Question 15

- (a) Given that $\sin(x + y) + \sin(x - y) = 2 \sin x \cos y$ 3

and $\cos(x + y) + \cos(x - y) = 2 \cos x \cos y$ and by using the fact that

$A + B + C = \pi$ for triangle ABC:

Show that $\frac{\sin A + \sin B}{\cos A + \cos B} = \cot \frac{C}{2}$

- (b) Let $f(x) = \begin{cases} \frac{\sin x}{x} & \text{for } 0 < x \leq \frac{\pi}{2} \\ 1 & \text{for } x = 0 \end{cases}$

- (i) Find the derivative of $f(x)$ for $0 < x < \frac{\pi}{2}$ and prove that $f'(x)$ is negative in this interval. 2

- (ii) Sketch the graph of $y = f(x)$ for $0 < x < \frac{\pi}{2}$ and deduce that $\sin x \geq \frac{2x}{\pi}$ in this interval 2

- (c) Let $\omega = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$. The complex number $\alpha = \omega + \omega^2 + \omega^4$ is a root of the quadratic equation $x^2 + ax + b = 0$, where a and b are real.

- (i) Prove that $1 + \omega + \omega^2 + \dots + \omega^6 = 0$. 2

- (ii) The second root of the quadratic equation is β . Express β in terms of positive powers of ω . Justify your answer. 1

- (iii) Find the values of the coefficients a and b . 2

- (iv) Deduce that $-\sin \frac{\pi}{7} + \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{2}$. 3

END OF EXAMINATON

Extention 2 Mathematics-2014 Half yearly solution

1. B 2. A 3. A 4. D 5. C 6. B 7. B 8. D 9. C 10. D	1 mark each	
Question 11 (15 marks) a) $3-2i = \sqrt{9+4} = \sqrt{13}$ b) $(3+4i)\overline{(4-i)} = (3+4i)(4+i)$ $= 12 + 3i + 16i - 4$ $= 8 + 19i$ c) (i) $\vec{AO} = i \vec{AB}$ $= i\alpha$ $\vec{OA} = -i\alpha$ correspond to vertex A. (ii) $\vec{OB} = \vec{AB} - \vec{OA}$ $= \alpha - i\alpha$ $= \alpha(1-i)$ corresponds to vertex B (iii) $A = \frac{1}{2} \times OA \times AB$ $= \frac{1}{2} \alpha ^2$ ($\triangle OAB$ is isosceles, $OA = AB$) $= \frac{1}{2} \alpha \cdot \alpha$ d) (i)	1 1 1 1 1 1 1	

$1 + \frac{1+i}{1-i} = \frac{1-i+1+i}{1-i}$ $= \frac{2(1+i)}{(1-i)(1+i)}$ $= 1+i$ $\therefore z = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$	1	
(ii) $z^5 = (\sqrt{2})^5 \operatorname{cis} \frac{5\pi}{4}$ $= 4\sqrt{2} \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i \right)$ $= -4 - 4i$	1	
e) Point of intersection: sub. $y = \sqrt{3}x$ into $x + \sqrt{3}y + 4 = 0$ $x + \sqrt{3} \cdot \sqrt{3}x + 4 = 0$ $4x = -4$ $x = -1$ and $y = -\sqrt{3}$ Radius = $\sqrt{3+1} = 2$ $m_1 = \tan^{-1}(\sqrt{3})$ $= \frac{\pi}{3}$ $m_2 = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ $= -\frac{5\pi}{6}$ $\therefore$ The conditions which satisfy the complex number z in the region are: $\operatorname{im} z \geq -1, \quad z + 1 + \sqrt{3} \leq 2 \quad \text{and} \quad \frac{\pi}{3} \leq \arg z \leq \frac{5\pi}{6}$	1 finding correct answer for point of intersection 1 Correct radius 1 correct gradients 2 for three correct conditions	

Question 12

(a) Let $y = x^3$, then $x = \sqrt[3]{y}$

Sub. x into $2x^3 - 2x^2 + 1 = 0$

$$2(\sqrt[3]{y})^3 - 2y^{\frac{2}{3}} + 1 = 0$$

$$2y + 1 = 2y^{\frac{2}{3}}$$

$$(2y + 1)^3 = 8y^2$$

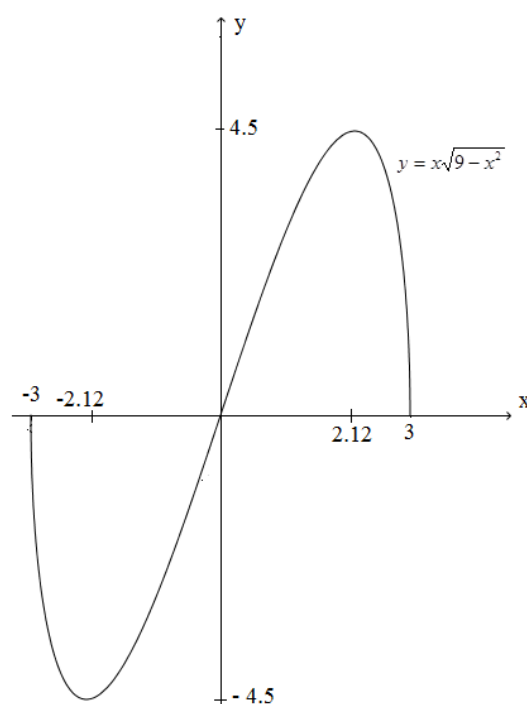
$$8y^3 + 12y^2 + 6y + 1 = 8y^2$$

$$8y^3 + 4y^2 + 6y + 1 = 0$$

$\therefore$ The equation with roots α^3, β^3 and γ^3 is

$$8x^3 + 4x^2 + 6x + 1 = 0$$

b)



(i)

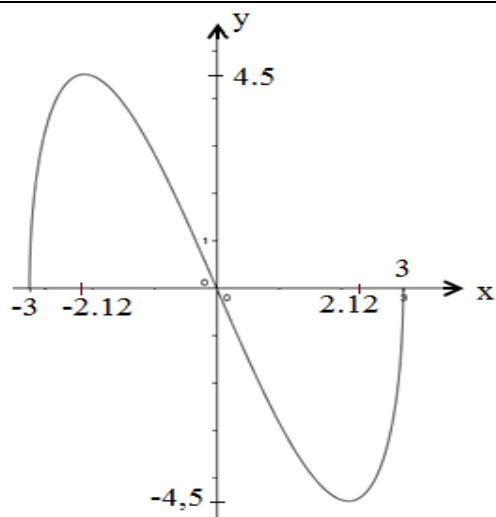
1
correct
substitution of x

1
simplifying

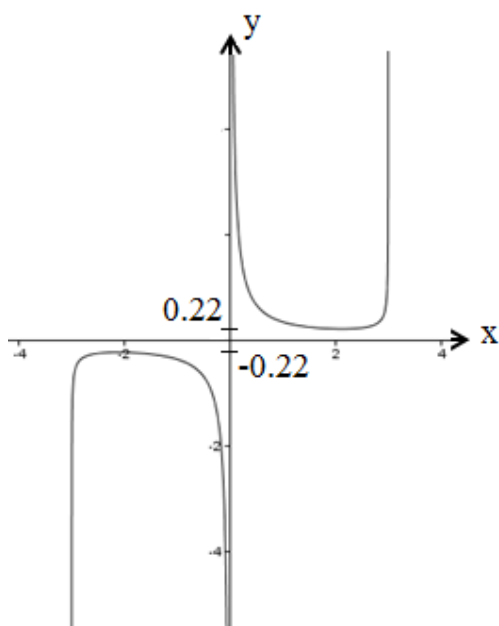
1
correct answer

1
both x -intercepts

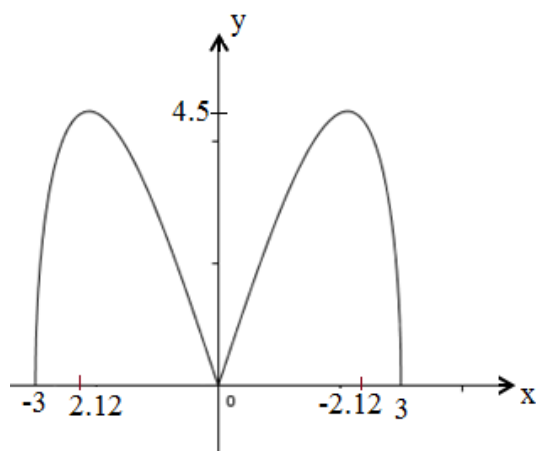
2
Correct values of
 x & y
coordinates
of max & min
points.



(ii)



(iii)



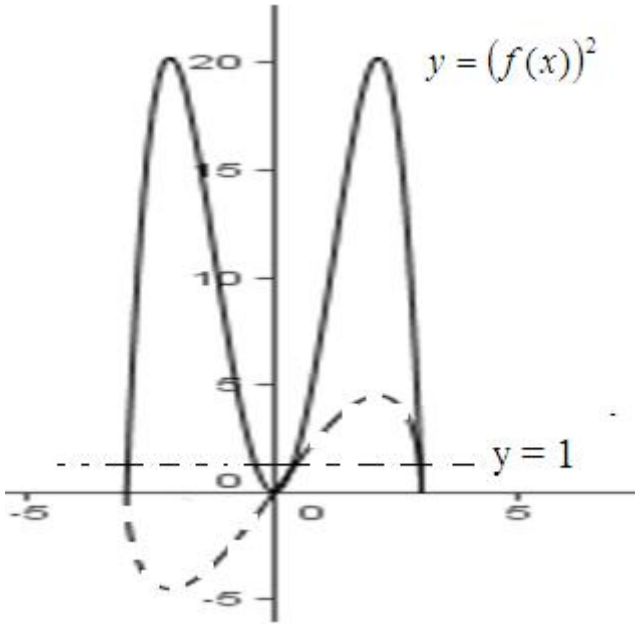
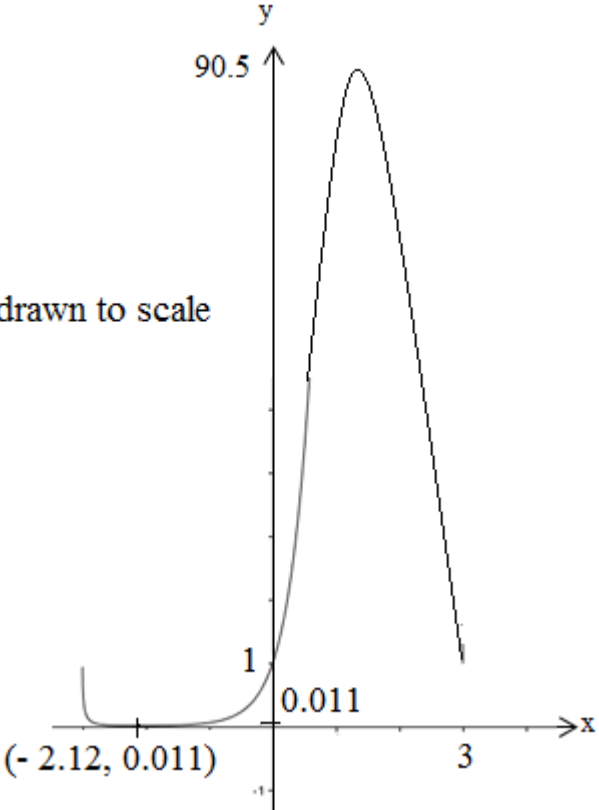
(iv)

1 correct shape
& locations of
max & min
points

1 correct shape

1
correct
approximation
values of max &
min points

1
correct shape &
values of max
points

 $y = (f(x))^2$	1 correct shape 1 The graph must be below original graph for $y < 1$	
(v) Not drawn to scale 	1 correct shape 1 correct approximation values of max & min points 1 y-coordinates of both end points of the graph is at 1	

Question 13 (15 marks)

asymptotes:

$$\text{Sub. } y = \pm \frac{\sqrt{5}}{2} x \text{ into (1)}$$

$$\frac{x \sec \theta}{2} - \frac{\sqrt{5} x \tan \theta}{2\sqrt{5}} = 1$$

$$\sqrt{5} x (\sec \theta - \tan \theta) = 2\sqrt{5}$$

$$x_1 = \frac{2}{\sec \theta - \tan \theta} \quad \text{similarly} \quad x_2 = \frac{2}{\sec \theta + \tan \theta}$$

Sub. x_1 into (1)

$$\frac{\sec \theta}{\sec \theta - \tan \theta} - \frac{y \tan \theta}{\sqrt{5}} = 1$$

$$\sqrt{5} \sec \theta - y \tan \theta (\sec \theta - \tan \theta) = \sqrt{5} (\sec \theta - \tan \theta)$$

$$y_1 = \frac{\sqrt{5}}{\sec \theta - \tan \theta} \quad \text{similarly} \quad y_2 = \frac{-\sqrt{5}}{\sec \theta + \tan \theta}$$

The mid-point of LM is:

$$\begin{aligned} M_x &= \frac{2}{2(\sec \theta - \tan \theta)} + \frac{2}{2(\sec \theta + \tan \theta)} \\ &= \frac{2(\sec \theta + \tan \theta) + 2(\sec \theta - \tan \theta)}{2(\sec^2 \theta - \tan^2 \theta)} \\ &= 2 \sec \theta \end{aligned}$$

$$\begin{aligned} M_y &= \frac{\sqrt{5}}{2(\sec \theta - \tan \theta)} - \frac{\sqrt{5}}{2(\sec \theta + \tan \theta)} \\ &= \frac{\sqrt{5}(\sec \theta + \tan \theta) - \sqrt{5}(\sec \theta - \tan \theta)}{2(\sec^2 \theta - \tan^2 \theta)} \\ &= \sqrt{5} \tan \theta \end{aligned}$$

The mid-point M is the coordinate of P

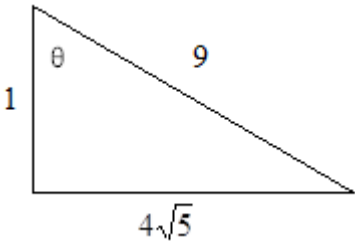
$$\therefore PM = LP$$

(iii)

1
finding x-
coordinates

1
Finding y
coordinates

2
finding the mid-
point coordinates of
LM
and conclusion

$OL = \sqrt{\frac{4}{(\sec \theta + \tan \theta)^2} + \frac{5}{(\sec \theta + \tan \theta)^2}}$ $= \frac{3}{\sec \theta + \tan \theta}$ Similarly $OM = \frac{3}{\sec \theta - \tan \theta}$ $m_{OL} = \frac{\sqrt{5}}{2} \quad \text{and} \quad m_{OM} = \frac{-\sqrt{5}}{2}$ $\tan \theta = \left \frac{\frac{\sqrt{5}}{2} - \frac{\sqrt{5}}{2}}{1 - \left(\frac{\sqrt{5}}{2}\right)^2} \right $ $= \frac{\sqrt{5}}{\frac{1}{4}} = 4\sqrt{5}$  $\text{Area } \triangle OLM = \frac{1}{2} OM \times OL \times \sin \theta$ $= \frac{1}{2} \times \frac{3}{\sec \theta - \tan \theta} \times \frac{3}{\sec \theta + \tan \theta} \times \frac{4\sqrt{5}}{9}$ $= 2\sqrt{5} \text{ units square}$ Which is independent of the position of P.	1 Finding the length OL & OM 1 finding the angle θ 1 finding $\sin \theta$ 1 finding the area and conclusion	
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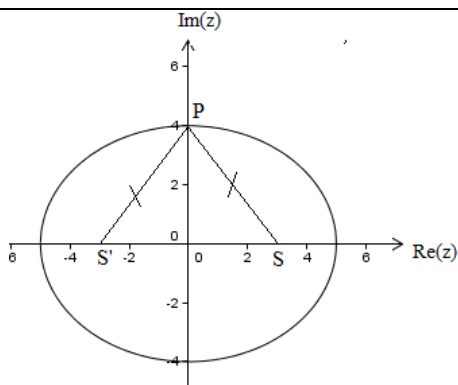
Question 14 (15 marks)

(a) $P(z)$ has real coefficient, its complex roots are in conjugate. $\therefore 2 + i$ is also a root. $(z - 2 + i)(z - 2 - i) = z^2 - 4z + 5$ By polynomial division or by inspection 2 and -1 are the other 2 roots. $\therefore P(z) = (z^2 - 4z + 5)(z^2 - z - 2)$ $= (z^2 - 4z + 5)(z - 2)(z + 1)$ (b) (i) $t_1 + t_2 + t_3 = 0$ (sum of roots) (ii) $t_1^2 + t_2^2 + t_3^2 = (t_1 + t_2 + t_3)^2 - 2(t_1t_2 + t_1t_3 + t_2t_3)$ $= 0 - 2 \times c$ $= -2c$ (iii) $\frac{df(t)}{dt} = 3t^2 + c = 0$ $\Rightarrow t = \pm \sqrt{-\frac{c}{3}}$ Let $\Rightarrow u = \sqrt{-\frac{c}{3}}$, $v = -\sqrt{-\frac{c}{3}}$ and $v = -u$ $f(u) \cdot f(v) = (u^3 + cu + d)(-u^3 - cv + d)$ $= \left(u(u^2 + c) + d\right)\left(-u(u^2 + c) + d\right) \quad (1)$ Since $3u^2 + c = 0$, sub $u^2 = -\frac{c}{3}$ into (1) $f(u) \cdot f(v) = \left(u\left(-\frac{c}{3} + c\right) + d\right)\left(-u\left(-\frac{2}{3} + c\right) + d\right)$ $= \left(u \cdot \frac{2c}{3} + d\right)\left(-u \cdot \frac{2c}{3} + d\right)$ $= d^2 - \frac{4u^2c^2}{9} \quad \text{Sub. } u = \sqrt{-\frac{c}{3}}$	1 state $2 + i$ is a root 1 Finding the factors 1 Correct answer 1 2 1 finding u and v 1 finding $f(u) \cdot f(v)$ 1 Showing sub. of $u^2 = -\frac{c}{3}$ 1 showing the	
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$\Rightarrow 1 + k + \frac{k^2}{2} + \frac{k^3}{5} = 1 + k + \frac{k^2}{2} + \frac{k^3}{5} + \frac{k^4}{24}$ $\Rightarrow \frac{k^4}{24} = 0$ $\therefore k = 0$ Sub. $k = 0$ into $E(x) \Rightarrow E(0) = 1$ $\therefore k$ is not a root of $E(x)$, by factor theorem hence $E(x)$ has no double root.	1 Conclusion with explanation	
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Question 15 (15 marks)

(a)		
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The graph of the complex number z is drawn above.

$$\begin{aligned} PS + PS' &= 2a \quad (\text{property of ellipse}) \\ &= 10 \end{aligned}$$

$$\therefore a = 5$$

By selecting P at the position shown,

$$\begin{aligned} OP^2 &= PS^2 - OS^2 \\ &= 25 - 9 = 16 \end{aligned}$$

$$\therefore b = 4$$

The Cartesian equation of the ellipse is

$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$

(b)

$$\text{Let } u = 2x + 3 \Rightarrow x = \frac{u-3}{2}$$

$$\text{and } du = 2 \, dx$$

$$\begin{aligned} \int \frac{x-1}{\sqrt{2x+3}} \, dx &= \frac{1}{4} \int \frac{u-5}{\sqrt{u}} \, du \\ &= \frac{1}{4} \int u^{\frac{1}{2}} - 5u^{-\frac{1}{2}} \, du \end{aligned}$$

$$= \frac{1}{4} \left(\frac{2}{3} u^{\frac{3}{2}} - 10u^{\frac{1}{2}} \right)$$

$$= \frac{1}{6} (2x+3)^{\frac{3}{2}} - \frac{5}{2} (2x+3)^{\frac{1}{2}} + C$$

(c) (i)

$$y = \frac{c^2}{x} \quad \frac{dy}{dx} = -\frac{c^2}{x^2}$$

1
correct value of a

1
correct value of b
and equation of
ellipse

1
correct
substitution

1
correct integration
and answer in
term of x

At $x = ct$ $\frac{dy}{dx} = -\frac{c^2}{c^2 t^2} = -\frac{1}{t^2}$ Equation of the tangent at T is: $y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$ $yt^2 - ct = -x + ct$ or $x + t^2 y = 2ct$ (ii) Similarly the equation of the tangent at Q and P are $x + p^2 y = 2cp \quad (1)$ $x + q^2 y = 2cq$ $y(p^2 - q^2) = 2c(p - q)$ $y = \frac{2c}{p + q} \quad \text{sub into (1)}$ $\Rightarrow x = 2cp - \frac{2cp^2}{p + q}$ $= -\frac{2cpq}{p + q}$ Since $pq = 1$ $x = \frac{2c}{p + q}$ $\therefore R\left(\frac{2c}{p + q}, \frac{2c}{p + q}\right)$ Hence the locus of R is the line $y = x$ Where $-c < x < 0$ and $0 < x < c$, since $pq = 1$. (c) (i) Area of the circle = πr^2 Area of the segment = $\frac{1}{2}r^2(\pi - 2\theta) - \frac{1}{2}r^2 \sin(\pi - 2\theta)$	1 finding 'm' 1 Correct equation of tangent 1 finding y co-ordinate 1 finding x coordinate 1 finding locus of R 1 Correct restrictions on the values of x. 2	
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$\frac{1}{2}r^2(\pi - 2\theta) - \frac{1}{2}r^2 \sin(\pi - 2\theta) = \frac{1}{3}\pi r^2$ $\frac{1}{2}r^2\pi - r^2\theta - \frac{1}{2}r^2 \sin 2\theta = \frac{1}{3}\pi r^2$ $\frac{1}{6}r^2\pi - r^2\theta = \frac{1}{2}r^2 \sin 2\theta$ $\therefore \sin 2\theta = \frac{\pi}{3} - 2\theta$ (ii) Both graphs shown only have one point of intersection. $\therefore \sin 2\theta = \frac{\pi}{3} - 2\theta \text{ has only one solution.}$	correct use of both formulae 1 correct answer 2 Correct graphs and explanation	
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Question 16 (15 marks)

(a) Let $A = x + y$ and $B = x - y$	1	
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$\therefore A + B = 2x \Rightarrow x = \frac{A+B}{2}$ $A - B = 2y \Rightarrow y = \frac{A-B}{2}$ $\therefore \sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$ and $\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$ $\frac{\sin A + \sin B}{\cos A + \cos B} = \frac{2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}}{2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}} \quad (1)$ $= \tan \frac{A+B}{2}$ Now $A + B + C = \pi$ $\therefore A + B = \pi - C$ $\therefore \tan \frac{A+B}{2} = \tan \frac{\pi - C}{2}$ $R.H.S = \tan \left(\frac{\pi}{2} - \frac{C}{2} \right)$ $= \cot \frac{C}{2}$ $= L.H.S \quad \text{from (1)}$ (b) (i) $f'(x) = \frac{x \cos x - \sin x}{x^2}$ $= \frac{x \cos x - \cos x \tan x}{x^2}$ $= \frac{\cos x (x - \tan x)}{x^2}$	expressing x and y in term of A & B 1 correct simplifying 1 Correct answer 1 Correct differentiation	
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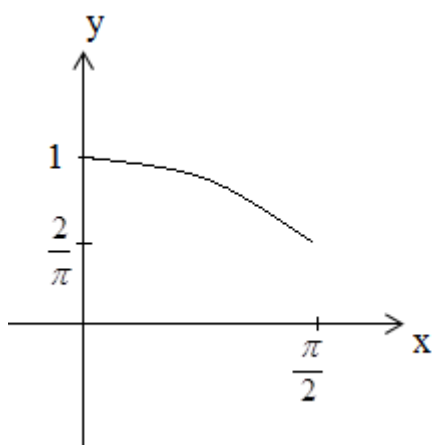
$$x^2 > 0, \cos x > 0 \text{ and } \tan x > x \text{ for } 0 < x < \frac{\pi}{2}$$

$$\therefore x - \tan x < 0, \text{ hence } f'(x) < 0$$

(ii)

$f(x)$ is a decreasing function for $0 < x < \frac{\pi}{2}$ as proven above.

$$f\left(\frac{\pi}{2}\right) = \frac{2}{\pi} \text{ and } f(0) = 1$$



From the graph $f(x) \geq \frac{2}{\pi}$ for $0 \leq x \leq \frac{\pi}{2}$

$$\text{ie } \frac{\sin x}{x} \geq \frac{2}{\pi}$$

$$\therefore \sin x \geq \frac{2x}{\pi}$$

(c) (i)

$$w = \text{cis } \frac{2\pi}{7}$$

$$\therefore w^7 = \left(\text{cis } \frac{2\pi}{7} \right)^7$$

$$= \text{cis } 2\pi$$

$$= 1$$

1
correct reason

1
correct graph

1
correct reason

1
expressing w^7 as $\text{cis } 2\pi$

$\therefore w$ is a root of $x^7 - 1 = 0$, that is of
 $(x-1)(x^6 + x^5 + \dots + 1) = 0$

Since $w \neq 1$, then w satisfy $x^6 + x^5 + \dots + 1 = 0$

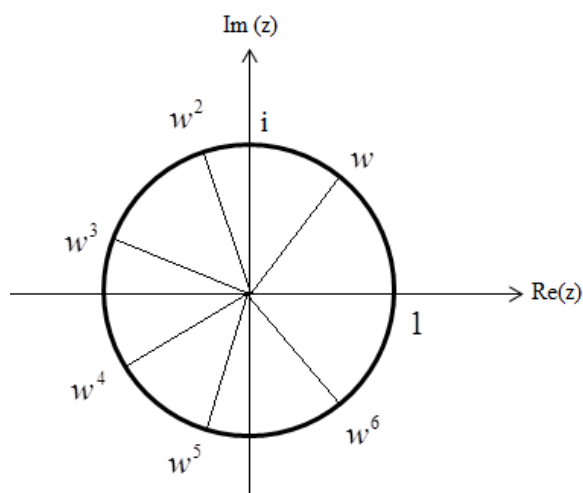
$$\therefore 1 + w + w^2 + \dots + w^6 = 0$$

(ii)

Since the quadratic equation $x^2 + ax + b = 0$ has real coefficients therefore complex roots are conjugate.

Hence $\bar{w}$ is also a root.

$$\therefore \beta = \bar{\alpha} = \bar{w} + \bar{w}^2 + \bar{w}^4$$



Since $\bar{w} = w^6$, $\bar{w}^2 = w^5$, $\bar{w}^4 = w^3$ from the diagram

$$\beta = w^3 + w^5 + w^6$$

(iii)

$$\alpha + \beta = -a$$

$$\Rightarrow a = -(w + w^2 + w^4 + w^3 + w^5 + w^6)$$

$$= -1 \quad (\text{using (i)})$$

$$= 1$$

1 explanation

1
correct answer with
explanation.

1 correct answer for
a

$\alpha\beta = b$ $= (w + w^2 + w^4).(w^3 + w^5 + w^6)$ $= w^4 (1 + w + w^3).(1 + w^2 + w^3)$ $= w^4 (1 + w^2 + w^3 + w + w^3 + w^4 + w^3 + w^5 + w^6)$ $= w^4 (1 + w + w^2 + + w^6 + 2w^3)$ $= w^4 (0 + 2w^3)$ $= 2w^7 = 2$ $\therefore \quad \quad \quad b = 2$	1 correct answer for b
(iv) Hence the quadratic equation is $x^2 + x + 2 = 0$ $x = \frac{-1 \pm \sqrt{1 - 4 \times 1 \times 2}}{2}$ $= \frac{-1 \pm i\sqrt{7}}{2}$ $\therefore \text{Im } \alpha = \pm \frac{\sqrt{7}}{2}$ Now, $\text{Im } \alpha = \text{Im}(w + w^2 + w^4)$ $= \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}$ $= \sin \frac{2\pi}{7} + \sin \left(\pi - \frac{3\pi}{7} \right) + \sin \left(\pi + \frac{\pi}{7} \right)$ $= -\sin \frac{\pi}{7} + \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7}$ $= \frac{\sqrt{7}}{2}$	1 Finding the roots of $x^2 + x + 2 = 0$ 1 expressing im α 1 simplifying to achieve the result